

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend JavaScript engineer with 6 years building production systems in **Node.js, MongoDB, and Express**. Led the end-to-end, zero-downtime migration of 20+ live applications from Parse/MongoDB to Firebase/GCP, reporting directly to the CTO — including a real-time two-way sync pipeline processing messages through Google Pub/Sub. Experienced across message queues (Kafka, Pub/Sub, NATS Streaming), **Kubernetes**, and network-driven systems (gRPC, USSD gateways, real-time data sync). Owns features end to end, from system design and architecture to production. Fully remote for 6 years across US and Singapore teams; fluent English.

CORE COMPETENCIES

Node.js Backend Development

MongoDB

Express.js

Kubernetes & Docker

Message Queues (Kafka · Pub/Sub · NATS)

System Design & Architecture

Real-Time Data Sync

React · HTML · CSS

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote · Reporting directly to the CTO

- Implemented **real-time two-way sync between MongoDB and Firestore** using Node.js and Google Pub/Sub for distributed message processing, keeping both datastores consistent under live production traffic
- Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime** — from architecture and system design through production cutover
- Saved the company **\$5,000/month** by migrating all services off AWS
- Implemented Express Shipping end to end — conception to production — generating an additional \$40 revenue per order
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion
- Migrated file storage from Amazon S3 to GCS via Python tooling

MTailor

Mar 2022 – Jul 2022

Backend Contractor — US Remote

- Built a migration pipeline reading from **MongoDB** and writing to Firestore after complex transformation logic
- Wrote engineering documentation for the new Firebase SDKs (Node.js and Python) and trained the Ops team on dashboards and backup procedures

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Led optimisation of a PWA delivering **real-time stock market prices**; built jobs querying 2M+ Firestore records against Morningstar APIs
- Extended backend APIs using **Express.js** and Firebase Cloud Functions hosted on Heroku
- Added responsive UI pages using HTML, Tailwind, and React

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backends for DeFi applications using Web3 and **Node.js**; delivered React UIs from Figma under tight timelines

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead — Uganda Remote

- Architected a **microservices backend using Apache Kafka, Docker, and Kubernetes** for FIDA Uganda's case management platform
- Built a USSD service for legal aid providers communicating with a Node.js backend via **gRPC**
- Led a team of 4 developers building legal tech systems for NGOs; shipped a cross-platform React Native app with Firebase push notifications

PROJECTS

Real-Time Pub/Sub Sync Pipeline

Live MongoDB ↔ Firestore two-way sync supporting parallel production traffic during a zero-downtime migration of 20+ apps.

Node.js · MongoDB · Google Pub/Sub · Firestore

Marketplace Recommendation Engine

Collaborative and content-based filtering engine built on MongoDB for service-provider matching.

Node.js · MongoDB

Healthcare WhatsApp Chatbot

Prescription OCR pipeline integrating OpenAI and Google Vision via the WhatsApp Business API.

Node.js · OpenAI · Google Vision · WhatsApp Business API

EDUCATION

BSc Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, MongoDB, Express, Firebase, GCP, React, Python, SQL, HTML, CSS, Git, Linux

Intermediate: Go, AWS, Rust, React Native **Infrastructure:** Kubernetes, Docker, Kafka, Pub/Sub, NATS Streaming, gRPC

Languages: English (fluent), Kiswahili